

CareerAI

1. Project Overview

CareerAI is an AI-powered placement preparation ecosystem built using **Gemini 3 Pro**. It unifies core placement modules like **DSA**, **Aptitude**, **Resume Analysis**, **Mock Tests**, **Job Matching**, and **AI Mock Interviews**, along with a real-time **AI Mentor Chatbot**.

The goal is to help students prepare faster, smarter, and more confidently for campus placements and software engineering roles.

2. Key Features

2.1 Visual DSA Pattern Library

- 22 core DSA patterns
 - Includes Sliding Window, Two Pointers, BFS/DFS, DP, Heaps, Backtracking
 - Pattern-based conceptual understanding
 - Clean UI with categorized problems
 - AI-powered explanations & code examples
-

2.2 AI Resume & ATS Analyzer

- Upload resume + job description
 - ATS Score Calculation
 - Missing keyword detection
 - Skill & role match analysis
 - AI suggestions to improve resume
 - AI job role recommendations
-

2.3 AI Mock Interview Coach

- HR and Technical interview simulation
 - Adaptive questioning (based on user's answers)
 - Supports multiple languages
 - HR Behavioral questions
 - Technical coding + CS fundamentals
 - Immediate AI feedback
-

2.4 Mock Test Engine

- Practice modes: 5, 10, 20, 30 questions
 - Company-specific tests (TCS, Infosys, Wipro, MAANG)
 - Coding + Aptitude mixed tests
 - Detailed score breakdown
 - Timer-based exam simulation
 - Instant explanations & solutions
-

2.5 Aptitude & Logical Reasoning Module

- Quantitative Topics
 - Logical Reasoning Topics
-

2.6 AI Power Tools Suite

- **Time Complexity Analyzer** – Detects Big-O complexity
 - **Pattern Predictor** – Identifies DSA pattern from description
 - **Code Dry Run Visualizer** – Step-by-step code execution
 - **Code Translator** – Converts code across languages
-

2.7 Ask Nik – AI Doubt Solver Chatbot

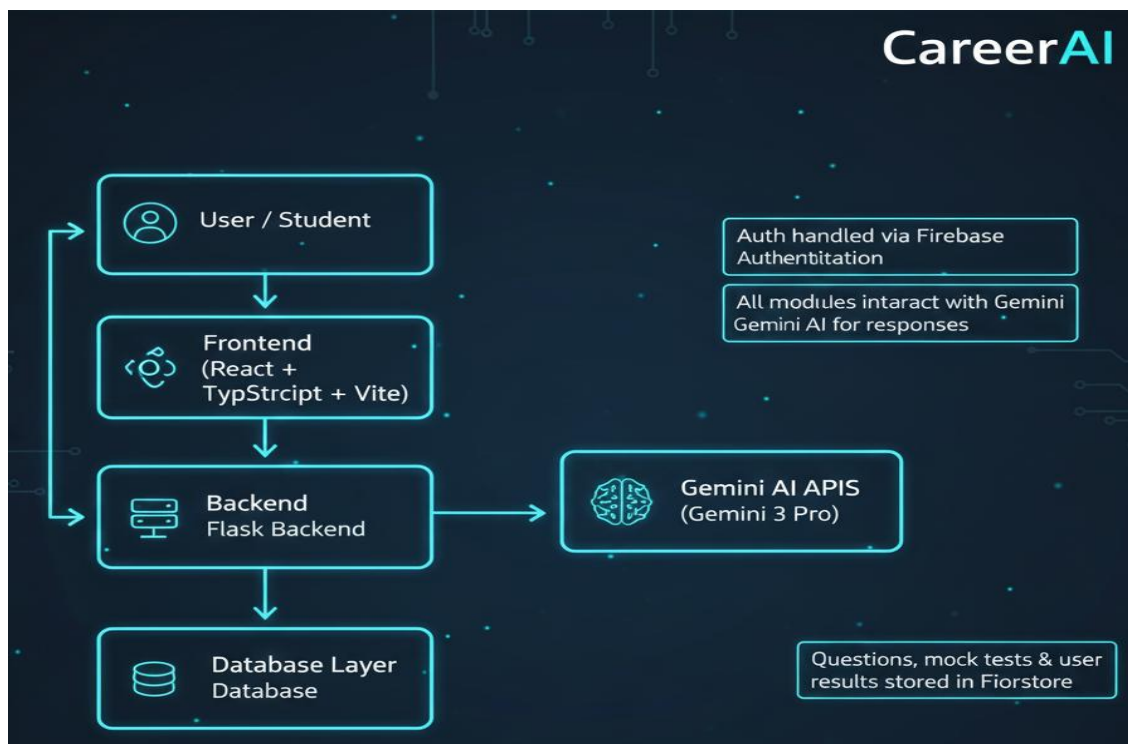
- Gemini 3 Pro powered assistant
- Explains DSA, interviews, aptitude, coding

- Debugs code & Generates examples and visual explanations
- 24x7 instant support & Creates personalized study plans

3. Tech Stack



4. System Architecture



5. Dataset & Question Banks

- Custom-built Aptitude Question Bank
 - Logical Reasoning Sets
 - Company-Specific Questions
 - JSON-Based MCQ Data
 - AI-generated coding problems
-

6. Deployment

- Web app deployed on Vercel
 - GitHub repository linked for CI/CD
 - Public accessible demo link included
-

7. Target Audience

- Engineering Students
 - Placement Aspirants
 - Competitive Programming Beginners
 - Job Seekers aiming for Software Roles
-

8. Problem Statement

Students struggle with:

- Scattered study resources
- Manual resume checking
- Lack of personalized interview practice
- Difficulty understanding DSA patterns
- Limited mock test platforms

CareerAI solves all these with one integrated system.

9. Impact

- Saves hours of preparation time
 - Improves resume quality significantly
 - Increases performance in DSA & aptitude
 - Boosts confidence in interviews
 - Provides personalized guidance to every user
-

10. Future Enhancements

- Mobile App version
 - Peer-to-peer mock interviews
 - Leaderboards & gamification
 - Advanced ML-based job recommender
 - Real-time coding editor for contests
-

11. Project Links

- **Live Website :** <https://career-ai-chi.vercel.app/>
- **GitHub Repository :** <https://github.com/NikhilShimpy/CareerAi>
- **Demo Video :** <https://youtu.be/hJ5Hinj04VQ?si=6dsjX51ypygCLyt3>

MEDIA GALLERY

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Mastery Suite

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DSA & APTITUDE

Pattern Library

Aptitude & Logic

Mock Test Engine

Power Tools

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Resume Analyzer

Job Search

Ask Nik

Developed By



Optimize Your Career

The ultimate AI-powered career platform. Master DSA patterns, analyze resumes, beat ATS systems, and crack interviews.

Learn Patterns

Check Resume

**DSA Mastery**
Master Sliding Window, Two Pointers, and 15+ patterns with visuals.
[Start Learning →](#)

**ATS Resume Scoring**
Get a detailed audit of your resume and fix missing keywords.
[Check Score →](#)

**AI Coach**
Practice HR and Technical rounds with an AI that mirrors your language.
[Start Mock Interview →](#)

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Pattern Library

Master these 22 patterns to crack any coding interview.

Search patterns...

**Sliding Window**
Perform operations on a specific window size of an array/list. Efficient f...
Max/Min sum of subarray size K
Longest substring with K distinct chars

**Two Pointers**
Use two pointers to iterate distinct indices, usually closing in or moving...
Sorted Arrays (Two Sum)
Palindrome checks

**Fast & Slow Pointers**
Hare & Tortoise algorithm. Use two pointers moving at different speeds.
Cycle in Linked List
Middle of Linked List

**Merge Intervals**
Dealing with overlapping intervals.
Scheduling
Meeting rooms

**Kadane's Algorithm**
Dynamic Programming approach to find maximum sum subarray.
Maximum Subarray Sum
Maximum Product Subarray

**Cyclic Sort**
Iterate array and place elements at correct indices.
Numbers in range 1 to N
Find missing/duplicate numbers

**In-place Reversal of LinkedList**
Reverse links between nodes without

**Breadth First Search (BFS)**
Level-by-level traversal.

**Depth First Search (DFS)**
Go deep before going wide.

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Sliding Window

Problem Set

DIFFICULTY: All

LeetCodeGFGHackerRankCodeChef

EASYLeetCode

Max Sum Subarray of Size K

Solve

MEDIUMLeetCode

Longest Substring Without Repeating Characters

Solve

HARDLeetCode

Minimum Window Substring

Solve

Load More Problems

PATTERN INSIGHT

Sliding Window

Perform operations on a specific window size of an array/list. Efficient for subarray/substring problems.

Core Intuition

Slide the window by adding one element and removing one, avoiding re-calculation.

When to use:

Max/Min sum of subarray size K

Longest substring with K distinct chars

String anagrams

VISUAL CONCEPT

[A B C] D E

→ Slide

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Aptitude & Logic

Master Quant shortcuts and Logical reasoning patterns.

AllQuantitativeLogical

Quantitative

Time and Work

Rate of work, efficiency, and man-days.

Quantitative

Pipes and Cisterns

Filling and emptying tanks.

Quantitative

Time, Speed, Distance

Relative speed, trains, boats.

Quantitative

Boats & Streams

Upstream and Downstream motion.

Quantitative

Profit and Loss

Gain, Loss, Discount, MP, CP, SP.

Quantitative

SI & CI

Interest calculations.

Quantitative

Percentages

Base of all arithmetic.

Quantitative

Ratio & Proportion

Comparison of quantities.

Quantitative

Mixture & Alligation

Mixing ingredients/prices.

Quantitative

Quantitative

Quantitative

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Back to Library

Time and Work

Quantitative

Rate of work, efficiency, and man-days.

TRICKS & SHORTCUTS

LCM Method: Total Work = LCM(Days)

M1D1H1 = M2D2H2

STRATEGY

Find unit work/day.

Add efficiencies.

VISUAL CONCEPT

$A(10d) + B(15d) \rightarrow LCM=30 \rightarrow A=3x$

Practice Problems

(6 loaded)

All Levels

Q1 EASY

A does work in 10 days, B in 15. Together?

Answer: 6 days

STEP-BY-STEP SOLUTION

Total=30. A=3, B=2. Total=5. $30/5 = 6$.

Q2 MEDIUM

A, B, and C can complete a piece of work in 10, 12, and 15 days respectively. They started working together. After 2 days, C left the work. In how many days will the remaining work be completed by A and B together?

Q3 MEDIUM

Pipe A can fill a tank in 10 hours, Pipe B can fill it in 15 hours, and Pipe C can empty it in 12 hours. If Pipe A is opened at 9 AM, Pipe B at 10 AM, and Pipe C at 11 AM, at what time will the tank be full?

Q4 MEDIUM

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Nikhil Shimpy

nikhilshimpyworking@g...

Test Generator Engine

Quick Practice

Full Mock Test

Company Specific

Focus Topic

Mixed (Quant + Logical)

Number of Questions

5 Qs

10 Qs

20 Qs

30 Qs

Generate Test & Start

QUESTION

1 / 5

7:28

Submit Test

Medium

Quantitative Aptitude

A train 150 meters long is running at a speed of 90 km/hr. How long will it take to cross a pole?

☐ 6 seconds

☐ 9 seconds

☐ 12 seconds

☐ 15 seconds

Previous

Next

Test Completed!

Here is how you performed

2

TOTAL SCORE

2

CORRECT

3

WRONG

40%

ACCURACY

Detailed Analysis

Take Another Test

📄

🏠

🧩

📊

📄

🔧

🕒

📄

🔍

🚗

👤

← Back to Summary

Question Analysis

Question 1

Wrong

A train 150 meters long is running at a speed of 90 km/hr. How long will it take to cross a pole?

Your Answer:
12 seconds

Correct Answer:
6 seconds

Explanation & Short Trick:

To cross a pole, the train needs to cover its own length. First, convert the speed from km/hr to m/s: $90 \text{ km/hr} * (5/18) = 25 \text{ m/s}$. Time = Distance / Speed = 150 meters / 25 m/s = 6 seconds.

Question 2

Wrong

Find the missing number in the series: 3, 7, 15, ___, 63, 127

Your Answer:
32

Correct Answer:
31

Explanation & Short Trick:

The pattern followed is $(\text{previous number} * 2) + 1$. $3 * 2 + 1 = 7$, $7 * 2 + 1 = 15$, $15 * 2 + 1 = 31$, $31 * 2 + 1 = 63$, $63 * 2 + 1 = 127$. So, the missing number is 31.

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Nikhil Shimpy

nikhilshimpyworking@g...

Power Tools

Time Complexity Calculator

Paste code to get Big O analysis.

Pattern Predictor

Paste a problem description to guess the pattern.

Dry Run Visualizer

Trace code execution step-by-step.

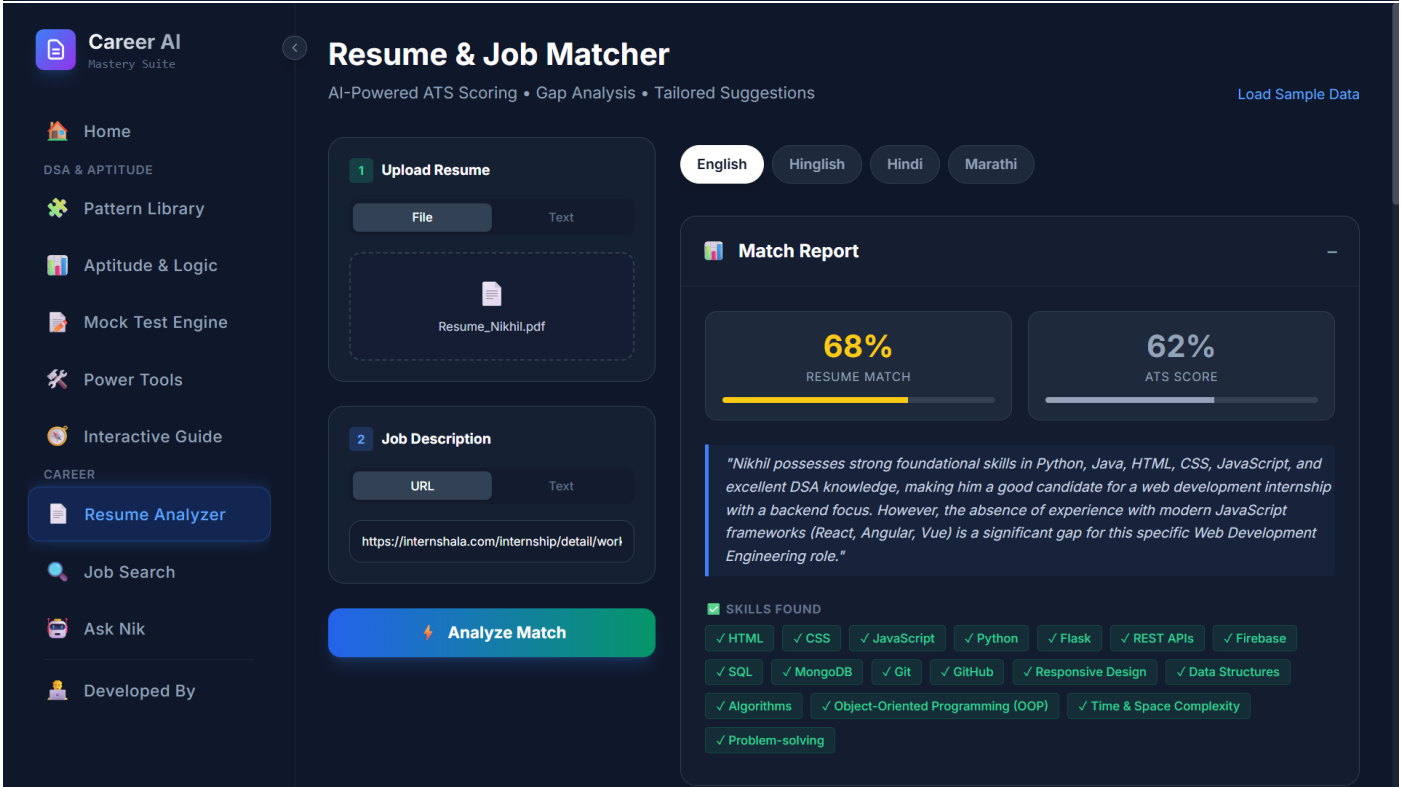
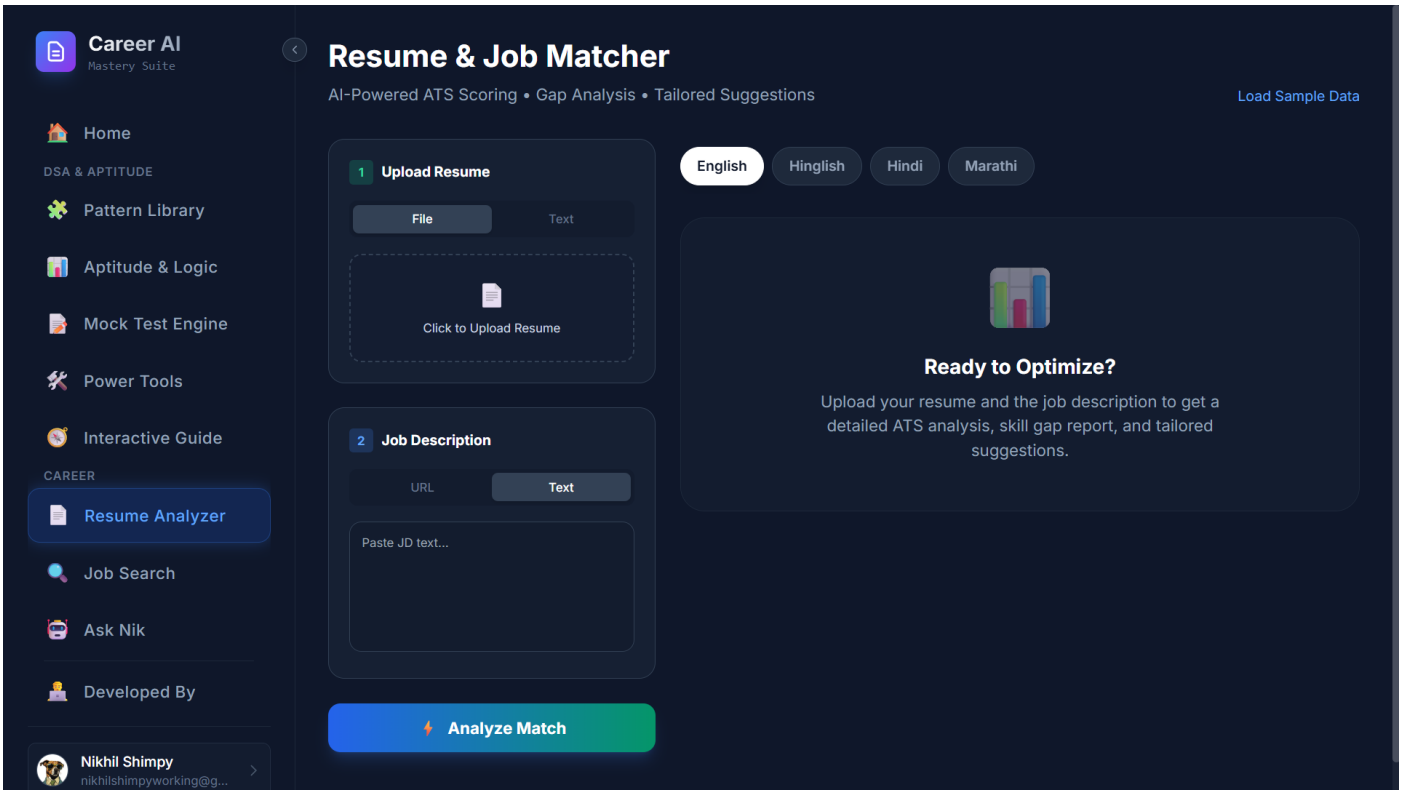
Code Translator

Convert code from one language to another.

Input for Time Complexity Calculator

Paste code or problem description here...

Run Analysis



https://internshala.com/internship/detail/work-from-home

Analyze Match

Recommended Jobs & Internships

Based on your resume skills and experience.

Backend Development Intern (Python)

98% Match

TechInnovate Solutions

Bengaluru, India

₹20,000 - ₹25,000 / month (Stipend)

3 days ago

Onsite

Python Django/Flask REST APIs SQL

View Job

Save

Full Stack Developer Intern (Python/Flask)

98% Match

NextGen Systems

Remote

\$800 - \$1200 / month (Stipend)

1 week ago

Remote

Python Flask JavaScript HTML/CSS

View Job

Save

Flutter Mobile Developer Intern

80% Match

AppGenius Labs

Hyderabad, India

₹18,000 - ₹22,000 / month (Stipend)

5 days ago

Hybrid

Flutter Dart Firebase REST APIs

View Job

Save

Backend Software Intern

98% Match

DataPulse Analytics

San Francisco, CA

\$20 - \$25 / hour (Stipend)

2 days ago

Onsite

Python AWS SQL Git

View Job

Save

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software engineer

Search

Show Filters

Showing 6 results

Sort: Recent

Software Engineer...

Onsite

✓ Verified

TechInnovate Solutions

Bengaluru, India

2-4 years

₹8-14 LPA

Java Spring Boot REST APIs SQL

3 days ago

Via Naukri →

Senior Software...

Hybrid

Login Req

Global Nexus Corp

Seattle, WA

5-8 years

\$140k-180k

Python AWS Microservices Kubernetes

1 week ago

Via LinkedIn →

Full Stack Software...

Remote

✓ Verified

StartupX Technologies

Remote (US)

2-5 years

\$90k-120k

React Node.js TypeScript MongoDB

2 days ago

Via Wellfound →

Software Development...

Onsite

✓ Verified

CloudSphere Solutions

Dublin, Ireland

4-7 years

€70k-95k

C# .NET Core Azure CI/CD

4 days ago

Via Indeed →

Staff Software...

Hybrid

Login Req

InnovateNow Inc.

San Francisco, CA

8+ years

\$190k-250k

Go Distributed Systems Kafka Docker

5 days ago

Via Glassdoor →

Frontend Software...

Onsite

Login Req

Creative Logic Ltd.

London, UK

2-5 years

£50k-70k

JavaScript React HTML/CSS Redux

1 day ago

Via LinkedIn →

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<

AI Problem Solver

Strict 12-Section Format Active

1. Pattern Detection

2. Visual Logic

3. Optimal Code (4 Languages)

4. Time Complexity Analysis

what is Array explain in very very short.

 **Array:** A collection of items of the **same data type**, stored at **contiguous memory locations**, and accessed using an **index**.

Paste your problem here...



Language Mirroring Active • 4-Language Code Gen

Clear Chat

AI Problem Solver

1. Pattern Detection
2. Visual Logic
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Nikhil Shimpy

AI & ML Enthusiast | Full Stack Developer

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[About](#)[QR](#)



About Me

Hi 🙌, I'm **Nikhil Shimpy** — a passionate IT student at SGSITS Indore (Batch 2027) with a deep love for building intelligent systems.

I thrive on solving real-world problems through innovative tech solutions. My

Education

B.Tech (IT)

SGSITS Indore
Batch 2027

Open to work?

YES

Nikhil Shimpy

 Portfolio LinkedIn GitHub

 Instagram

✦ Twitter

About

QR

About Me

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Nikhil Shimpy

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December 13, 2025



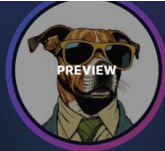
LAST LOGIN

December 13, 2025



AUTH METHOD

Email



PREVIEW

DISPLAY NAME

Nikhil Shimpy

1 Select an Avatar



SELECTED



Save Profile

Cancel